Shadhini JAYATILAKE

Software Engineer | BSc. Eng. (Hons.) | 5 years of Experience

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Date of Birth : 23-03-1994

With a fervent enthusiasm for software engineering, I bring a proven track record of achievements in Open Banking, Blockchain, Automation, and Algorithm Development. Over the past five years, I have actively contributed to the dynamic realms of finance, weather forecasting, and disaster risk reduction. My work showcases a harmonious blend of technical expertise and a curiosity-driven approach to innovation, coupled with a fast-learning mindset. This ability to quickly grasp new concepts enhances my adaptability and contributes to my success in diverse projects.

EDUCATION

2023 - Present — University of Moratuwa
MSc in Computer Science and Engineering - Software Architecture Specialization
Domain of ongoing Research : Blockchain, Weather Data

2014 - 2018 — University of Moratuwa
BSc. Eng. (Hons) in Computer Science and Engineering
Second Class - Upper Division
Dean's list - Semester 8 (SGPA : 4.05) | Overall GPA : 3.50

2000 - 2013 — Maliyadeva Girls' College, Kurunegala
G.C.E. Advanced Level, August 2013 – Physical Science Stream
Physics : A | Mathematics : A | Chemistry : B | Z-score : 1.9899 | Island rank : 698
G.C.E. Ordinary Level, December 2010 - All A's

EXPERIENCE

July 2023 - Present — **Software Engineer, FREELANCER, Sri Lanka**

- Understanding Customer Requirements, Solution Design and Proposal
- Develop and demonstrate PoCs (Proof of Concepta) to showcase the capabilities of WSO2/other solutions to potential clients.
- Guide and assist the WSO2 Services team in deployments, development processes, and issue troubleshooting to help overcome client's challenges.
- Contribution to Product Development

Java | WSO2 Identity Server | WSO2 API Manager | Open AI Library | Large Language Models (LLM) | Java | Java | Open AI GPT-3 | Python | Django | Java Spring Hibernate

October 2020 - June 2023 — **Software Engineer, WSO2, Sri Lanka**

- Middle-ware for Web Services : Identity and Access Management, API Management, Streaming Integration and Analytics
- Financial Solutions
  - PSD2/ CDR/ KSA Compliant Open Banking Solutions
- Led the Open Banking (OB) toolkit implementation and delivery for the Saudi Arabian Monetary Agency and provided continuous assistance for the Gulf International Bank (GIB) Compliance effort.
- Led end-to-end feature development, bug fixes, and enhancements for the OB solution, while researching technical improvements, introducing design enhancements, leading code/design reviews, and ensuring adherence to coding best practices and integration testing, with expertise in WSO2 base products, Gateway, MicroGateway and Analytics
- Supervised and guided interns/projects, broke down large tasks, delegated responsibilities, collaborated on development timelines, and managed ad-hoc tasks to support team efforts during feature releases, customer demands, and support incidents.
- Contributed to test automation and integration efforts.
- Pre-sales activities : Product demos and partner training programs
- Customer Support : Issue resolution, timely communication, and patch delivery
- Collaboration : Engaged with various extended teams to address urgent customer needs
- Customer Focus : Emphasized compliance and feature-related requirements
- Communication : Ensured comprehensive adherence to effective open-source communication practices, including timely communication of assumptions, blockers, and documentation of design decisions, architecture diagrams, while actively engaging with the Open Banking community through regulatory meetings and support portals

Java8 & Java11 | WSO2 Open Banking | Siddhi | Linux | WSO2 Identity Server | WSO2 API Manager | WSO2 Stream Processor | Open ID Connect | OAuth2.0 | FAPI | MTLS | Docker | CDS | OBIE-PSD2 | SAMA-KSA | Microservices Architecture | RESTful APIs

February 2019 - October 2020

Research Engineer, Center for Urban Water, Sri Lanka

- Data Engineering :
  - Redesigning relational databases for geospatial time series data.
  - Efficient and effective data routine implementation and maintenance. Software implementation for retrieving, extracting, processing, and validating time series data.
- End-to-end workflow automation for data.
- Optimization of algorithms and multi-core concurrent processing to improve process execution efficiency.
- Web Scraping and Data Gathering
- Research : Information collaboration platform for weather forecasting; investigation into how to construct a weather data collaboration platform to promote collaborative decision making.

MySQL | Python | Web Sraping | Google Cloud Platform | NetCDF | Blockchain | Linux

June 2017 - December 2017

Undergraduate Trainee | IFS R&D International (Pvt.) Ltd, Technology Team, Platform and Cloud Team

- Web Application Development
  - OData to Open API Converter : A web-based utility that includes components for OData metadata parsing, data conversion, and OpenAPI metadata serialization.
- Research on how to implement an authentication framework for the IFS framework that allows them to authenticate end users to IFS applications using the OpenID Connect identity layer, with the support of external identity providers rather than their own user store.
- Expertise in software development with industry-standard technologies.
- Professionalism, presentation skills, and technical writing abilities.

OData | Open API | OAuth2.0 | OpenID Connect | WSO2 Identity Server | Apache Tomcat | IntelliJ Idea | Maven | Java8 | XSLT | JSON | JWT

Publications

2020 Tharangana Perera, Sithija Thewa Hettige, Shadhini A. P. Jayatilake, Wickramaranga Abeygunawardhana and H. M. N. Dilum Bandara
Blockchain-Based Product Traceability Beyond Proof of Existence
2020 Moratuwa Engineering Research Conference (MERCon), Moratuwa, Sri Lanka,
2020, pp. 1-6, doi : 10.1109/MERCon50084.2020.9185207.

Technical Skills

Technical Fields	Software/Web Development, Open Banking, Workflow Automation, Distributed Systems, Blockchain, Machine Learning
Programming	Java, Python, Siddhi, HTML, CSS
Databases	MySQL, Oracle, Postgres, SQLite, H2
APIs, Frameworks	Python (Django, Pandas, Geopandas, Scrapy, Gcloud, Multiprocessing), Scikit Learn
Architectures	Microservices Architecture, RESTful APIs
WSO2 Middleware	WSO2 Identity Server, WSO2 API Manager, WSO2 API Manager Analytics
Specifications/ Standards	FAPI2, OIDC, Auth2.0, MTLS, OpenAPI (Swagger), OData Open Banking (CDR/ PSD2/ KSA)
Platforms	AWS, Hyperledger Fabric, Metabase, Linux, MacOS, Windows
Development Environments, Tools, IDEs	Docker, Maven, Git, Gradle, TestNG, JMeter, Postman, JIRA, Jenkins, Service Now, IntelliJ IDEA Ultimate Edition, IntelliJ IDEA PyCharm, Anaconda Cloud, Netbeans, GoLand, Web Storm, Visual Studio Code

Projects

Financial Assistant ChatBot June 2023 - July 2023

github.com/shadhini/RAD_financial_ai_chat_bot

The ChatBot powered by Generative text API and NLP, provides personalized financial advice, engages in intelligent conversations, supports queries on Sri Lankan investment options, offers multilingual support, and generates comprehensive reports, empowering users with informed decision-making and wealth management.

Python | Python Gradio Library | Open AI Library | Large Language Models (LLM) | Open AI GPT-3

IntelliTestSuite : Prototype Implementation of General-Purpose Testing Engine — June 2023 - July 2023

github.com/shadhini/AutoTestGen

A prototype application enhances software development with two main features : Test Case Generation, utilizing OpenAI GPT-3 to automate test model creation, scenarios, and data, and the Code Quality Checker, offering feedback on code flaws and improvements, to streamline testing and code review processes.

Python | Django | Open AI Library | Large Language Models (LLM) | Open AI GPT-3

Lyrics Genre Classifier & Visualization System — Mar 2023 - Apr 2023

github.com/shadhini/spark-ml-samples/tree/lyrics_classification_cherry_picked

Leveraging MLlib, this system achieves accurate song categorization by merging datasets and retraining models. Its web interface enables users to input lyrics and visualize predicted genre distribution intuitively.

MLlib (Machine Learning library for Apache Spark) | Apache Spark | Java | Spring Boot | Gradle

Comparative Analysis of MapReduce and Apache Spark for Airline Delay Data Processing — Mar 2023 - Apr 2023

github.com/shadhini/UoM_MapReduce-vs-Spark

This project compares MapReduce and Apache Spark for processing airline delay data using Hadoop on an AWS EMR cluster and demonstrates the results.

Apache Spark | Scala | Spark SQL | Hadoop | MapReduce | Hive | AWS EMR (Elastic MapReduce) | AWS S3 (Simple Storage Service)

CUrW Data Frameworks for Model Simulation and Event Simulations — FEB 2019 - JUNE 2020

github.com/CUrW-SL/curw_sim_db_utils github.com/CUrW-SL/event_sim_utils

A data framework that is responsible for populating a database with processed data, utilizing various spatial timeseries data distribution and calculation methods. This data serves as the input for model executions.

Python | Bash | Cronjobs | MySQL | Pandas | Scrapy

CUrW Data Managers — FEB 2019 - JUNE 2020

github.com/CUrW-SL/curw_flo2d_data_manager github.com/CUrW-SL/curw_mike_data_handler
github.com/CUrW-SL/curw_wrf_data_pusher github.com/shadhini/curw_db_adapter

Data Managers for HECHMS, FLO2D and MIKE models which prepares all kind of input files needed to run a simulation and then extract simulation outputs.

Python | Bash | NetCDF | MySQL | Pandas

CUrW Database Adapter — FEB 2019 - JUNE 2020

github.com/shadhini/curw_db_adapter

Database adapter, a library for accessing CUrW databases and executing operations on them

Python | MySQL | pymysql

BLOCKCHAIN-BASED PRODUCT TRACEABILITY | FINAL YEAR RESEARCH PROJECT — MAR 2018 - DEC 2018

Role : a member of a group of 4

A supply chain product tracking system which is capable of maintaining supply chain linkage, validating invariants, and supporting public auditability within the blockchain itself while reducing the cost, and enhancing privacy.

Hyperledger Fabric | Distributed Systems | PKI (Public Key Infrastructure) | Level DB | Node.js

DengAI : PREDICTING DISEASE SPREAD — MAY 2018 - JUL 2018

World rank by July 2018 : 92 | Role : a member of a group of 4

A ML(Machine Learning) model developed for the "DengAI" competition hosted by DrivenData Foundation which can predict the total number of dengue cases based on environmental variables describing changes in temperature, precipitation, vegetation, and more, for a given (city, year, week-of-year) data tuple.

Python | Scikit-learn | Pandas

ODATA TO OPENAPI CONVERTER | INTERNSHIP PROJECT — SEP 2017 - NOV 2017

Individual Project

A web tool which generates an interactive GUI to explore capabilities of a given OData service.

OData | XSLT | Swagger (Open API) | Java EE | Maven | Tomcat

Licenses & Certifications

2023 **Kaggle : Feature Engineering** - Certificate of Completion
2023 **AWS Hadoop Fundamentals** - Course Completion Certificate
2023 **AWS : Data Analytics Fundamentals** - Course Completion Certificate
2023 **AWS : Getting Started with Amazon DocumentDB (with MongoDB compatibility)** - Course Completion Certificate
2023 **AWS : Introduction to Amazon Elastic MapReduce (EMR)** - Course Completion Certificate

Awards & Achievements

2023 **Kaggle Competition : House Prices - Advanced Regression Techniques** - Ranked 317 in the world as of July 2023
2018 **DengAI : Predicting Disease Spread** - Ranked 92nd in the world as of July 2018
2017 **IEEEXTREME 11.0 Coding Competition** - World rank : 390 | Island rank:33
2013 **Sri Lankan Physics Olympiad** - Silver Medallist
2013 **Australian National Chemistry Quiz** - Award of Distinction

Extra-Curricular Activities

2011 - 2012 The Secretary of the Buddhist Association of Maliyadeva Girls' College
2006 All Island Western Music and Dance Competition - Winners | Category : Accordion Group
2004 All Island Western Music and Dance Competition - Runners Up | Category : Percussion Band
2009 Grade 4 Piano | ABRSM (Associated Board of the Royal Schools of Music) - Passed with a Merit
2009 Grade 5 Theory of Music | ABRSM – Passed with a Distinction
2007 Vice-secretary of the Buddhist Association

Non Technical Skills

- Problem Solving
- Project Management
- Teamwork
- Responsibility
- Presentation skills
- Self-Motivated

Interests

- Software/Web Development
- Distributed Systems
- Open Banking
- Blockchain Technology
- Machine Learning

Languages

Sinhala ●●●●●
English ●●●●●

References

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